

Exam Date & Time: 17-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: HUM 3021/HUM_3021 ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT
(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT [HUM 3021]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Mr. Arjun, aged 35, decides to follow stricter financial discipline while ensuring he spends quality time with his family. He plans to invest ₹15,000 at the end of every month in an investment account until he turns 50 years old (a total of 180 payments).

For the first five years (from age 35 to 40), the investment earns 20% nominal interest compounded quarterly. After the fifth year, the return is expected to reduce to 10% per annum, as he shifts to safer investment options.

Beginning at the end of his 40th year, Arjun starts to withdraw ₹2,40,000 every year to fund family vacations, while continuing his regular monthly investments until he turns 50. The final withdrawal is also made at the end of his 50th year.

Assuming all deposits and withdrawals occur at the end of their respective periods, determine the total amount that will remain in Arjun's account immediately after the withdrawal at age 50.

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>	
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	(4)
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	

- 1B) A manufacturing company is evaluating several investment opportunities to improve its operations. The available proposals fall under two main categories - Energy Efficiency Upgrades (E-series) and Production Equipment Expansion (P-series). Each project offers different annual returns and requires a specific initial investment. (4)

Proposal	Description	First Cost	Net Annual Income (year1-8)
E1	LED Lighting and HVAC Automation	1,20,000	24,000
E2	Solar Roof Panel Installation	1,50,000	28,000
P1	CNC Machine Upgrade	2,20,000	35,000
P2	New Packaging Line	3,20,000	50,000

The company has a total of ₹4,00,000 available for investment. It may select either one proposal within a category or a combination of proposals from different categories, provided that the total investment

does not exceed ₹4,00,000.

Using the Present Worth (PW) Method at an interest rate of 12% per year. Determine which investment or combination of investments provides the most economically advantageous outcome over an 8-year project life?

12%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>	<i>P/G</i>
1	1.120	.8929	1.0000	1.1200	1.000	0.893	0	0
2	1.254	.7972	.4717	.5917	2.120	1.690	0.472	0.797
3	1.405	.7118	.2963	.4163	3.374	2.402	0.925	2.221
4	1.574	.6355	.2092	.3292	4.779	3.037	1.359	4.127
5	1.762	.5674	.1574	.2774	6.353	3.605	1.775	6.397
6	1.974	.5066	.1232	.2432	8.115	4.111	2.172	8.930
7	2.211	.4523	.0991	.2191	10.089	4.564	2.551	11.644
8	2.476	.4039	.0813	.2013	12.300	4.968	2.913	14.471
9	2.773	.3606	.0677	.1877	14.776	5.328	3.257	17.356
10	3.106	.3220	.0570	.1770	17.549	5.650	3.585	20.254

- 1C) A manufacturing company incurred an initial loss of ₹10,00,000 due to environmental non-compliance. With corrective actions and improved pollution control systems, the loss amount is expected to be reduced by a fixed amount each year over the next five years. The total estimated future worth of loss over this five-year period is ₹35,00,000. Assuming that the reduction in losses is uniform (i.e., the same each year) and that the company uses a discount rate of 10% per annum to evaluate the time value of money, determine the annual reduction in loss value.

10%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>	<i>P/G</i>
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891

- 2A) A Ltd. depreciates its equipment at 10% p.a. on declining balance method. On 2019 the undepreciated amount of the equipment was ₹8,50,000 (Original Cost ₹12,00,000). Determine the book value of the asset in the years 2022 and 2023. Also, in 2025 if the machine is expected to be sold for ₹2,30,000 what will be its sunk loss or capital gain. (4)
- 2B) Your company needs to choose between two punch press models for purchase. The financial details for each are summarised below, and both machines have a 5-year lifespan with a residual scrap value of \$15,000. Using a 12% interest rate, perform a present worth analysis for a study period of four years to determine the more economical option. (3)
- Model A:** Initial cost is \$250,000. It has an annual operating cost of \$4,000 and generates an annual benefit of \$89,000.
- Model B:** Initial cost is \$205,000. It has an annual operating cost of \$4,300 and generates an annual benefit of \$86,000.
- Assume the market value of the asset is equal to the salvage value and the company adopts straight line

method of depreciation.

12%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>	<i>P/G</i>
1	1.120	.8929	1.0000	1.1200	1.000	0.893	0	0
2	1.254	.7972	.4717	.5917	2.120	1.690	0.472	0.797
3	1.405	.7118	.2963	.4163	3.374	2.402	0.925	2.221
4	1.574	.6355	.2092	.3292	4.779	3.037	1.359	4.127
5	1.762	.5674	.1574	.2774	6.353	3.605	1.775	6.397
6	1.974	.5066	.1232	.2432	8.115	4.111	2.172	8.930
7	2.211	.4523	.0991	.2191	10.089	4.564	2.551	11.644
8	2.476	.4039	.0813	.2013	12.300	4.968	2.913	14.471
9	2.773	.3606	.0677	.1877	14.776	5.328	3.257	17.356
10	3.106	.3220	.0570	.1770	17.549	5.650	3.585	20.254

- 2C) The Harmony Constructions Pvt. Ltd. is bidding for a government housing project aimed at constructing low-cost homes for needy families in a suburban region. As per the tender conditions, the company must design, construct, and maintain the housing colony for the next 100 years (assume $n=\infty$). The project engineers have estimated the following details: The initial cost of design, materials, and construction is estimated at ₹6,50,00,000. The cost of developing essential infrastructure such as roads, drainage, and water connections within the colony is estimated at ₹75,00,000. The annual maintenance cost (for building repairs, painting, and common area maintenance) is expected to be ₹8,00,000 in the first year and will increase by ₹80,000 every year due to ageing of structures and inflation. After 10 years, a major renovation (including roof replacement and repainting) will be required, costing ₹60,00,000, and it will recur every 10 years thereafter. After 15 years, the company plans to adopt a new maintenance material technology at a cost of ₹40,00,000, which will reduce the annual maintenance cost to ₹3,00,000 per year, with no further increase. Assume a MARR of 8% per annum and determine the total amount Harmony Constructions Pvt. Ltd. should quote in its proposal.

8%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.080	.9259	1.0000	1.0800	1.000	0.926	0
2	1.166	.8573	.4808	.5608	2.080	1.783	0.481
3	1.260	.7938	.3080	.3880	3.246	2.577	0.949
4	1.360	.7350	.2219	.3019	4.506	3.312	1.404
5	1.469	.6806	.1705	.2505	5.867	3.993	1.846
6	1.587	.6302	.1363	.2163	7.336	4.623	2.276
7	1.714	.5835	.1121	.1921	8.923	5.206	2.694
8	1.851	.5403	.0940	.1740	10.637	5.747	3.099
9	1.999	.5002	.0801	.1601	12.488	6.247	3.491
10	2.159	.4632	.0690	.1490	14.487	6.710	3.871
11	2.332	.4289	.0601	.1401	16.645	7.139	4.240
12	2.518	.3971	.0527	.1327	18.977	7.536	4.596
13	2.720	.3677	.0465	.1265	21.495	7.904	4.940
14	2.937	.3405	.0413	.1213	24.215	8.244	5.273
15	3.172	.3152	.0368	.1168	27.152	8.559	5.594

- 3A) The Ascarate Fishing Club, a nonprofit organization that teaches kids how to fish, is planning to provide a pond stocked with fish for children who have never caught a fish before. The club is considering two options. Option 1 is an above-ground vinyl pool costing ₹3,32,000, which incurs ₹5,000 maintenance per event. The pool is expected to last 10 years. Option 2 is an in-ground pond that will be 15 feet in diameter and 3 feet deep, dug by club members at no cost. The pond liner costs ₹830 per square foot, requiring 300 ft² for a total liner cost of ₹2,49,000, and a chain-link fence around the pond will cost ₹8,30,000. The pond also incurs a variable maintenance cost of ₹1,000 per event, in addition to its annual maintenance cost of ₹16,600. The pond is expected to be used for 10 years. The interest rate is 8% per year. Determine the number of events per year the club

must organize in the above-ground pool so that the total costs of the two options are equivalent.

8%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.080	.9259	1.0000	1.0800	1.000	0.926	0
2	1.166	.8573	.4808	.5608	2.080	1.783	0.481
3	1.260	.7938	.3080	.3880	3.246	2.577	0.949
4	1.360	.7350	.2219	.3019	4.506	3.312	1.404
5	1.469	.6806	.1705	.2505	5.867	3.993	1.846
6	1.587	.6302	.1363	.2163	7.336	4.623	2.276
7	1.714	.5835	.1121	.1921	8.923	5.206	2.694
8	1.851	.5403	.0940	.1740	10.637	5.747	3.099
9	1.999	.5002	.0801	.1601	12.488	6.247	3.491
10	2.159	.4632	.0690	.1490	14.487	6.710	3.871

- 3B) A commercial laundry service is planning to purchase a new washing machine. The supplier has proposed two models - Washing Machine A and Washing Machine B, both capable of handling similar workloads but differing in cost and operating efficiency. The finance manager must decide which machine to buy. The details of the two machines are as follows:

Particulars	Washing Machine A	Washing Machine B
Initial cost,	₹1,00,000	₹1,62,000
Estimated Life	12 years	12 years
Annual Operating costs	₹60,000	₹50,000
Salvage Value	Nil	₹27,982

(4)

The company's minimum attractive rate of return (MARR) is 15%. Using the Incremental Internal Rate of Return (IRR) method, determine the incremental IRR and then recommend which washing machine - A or B - should be selected.

- 3C) Britannia Industries is planning to borrow funds to expand its production facilities. The company has received three loan offers from different suppliers/vendors:
- Supplier 1: 13% compounded semi-annually
- Supplier 2: 6% per semi-annual compounded quarterly
- Supplier 3: 9% per annum, compounded monthly
- Britannia will make **equal quarterly payments**. Evaluate the effective rate on the payment period and determine which supplier provides the most cost-effective loan.

(2)

- 4A) A car purchased six years ago can be sold now for 3.5 lakhs or can be kept for another five years. In that case, it will have an annual operation and maintenance cost of Rs. 50,000 for the first year of operation from now and increases every year by an amount of 20,000. It is expected to have a salvage value of Rs. 2.25 Lakhs, Rs. 2 Lakhs, Rs. 1.75 Lakhs, Rs. 1.5 Lakhs, Rs. 1.25 Lakhs for years seven to eleven respectively. A suitable replacement is priced at Rs. 8 Lakhs and it has an annual operation and maintenance cost of Rs. 30,000. The salvage value of the car is Rs. 5 Lakhs after six years. Using outsider's point of view approach, examine whether it is financially viable to

(4)

make replacement at an interest rate of 8%.

8%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.080	.9259	1.0000	1.0800	1.000	0.926	0
2	1.166	.8573	.4808	.5608	2.080	1.783	0.481
3	1.260	.7938	.3080	.3880	3.246	2.577	0.949
4	1.360	.7350	.2219	.3019	4.506	3.312	1.404
5	1.469	.6806	.1705	.2505	5.867	3.993	1.846
6	1.587	.6302	.1363	.2163	7.336	4.623	2.276
7	1.714	.5835	.1121	.1921	8.923	5.206	2.694
8	1.851	.5403	.0940	.1740	10.637	5.747	3.099
9	1.999	.5002	.0801	.1601	12.488	6.247	3.491
10	2.159	.4632	.0690	.1490	14.487	6.710	3.871

4B) The details of Shreenath Company are as under:

Sales (40% cash sales)	₹15,00,000
Less: Cost of sales =	₹7,50,000
Gross Profit	₹7,50,000
Less: Office Exp. (including int. on debentures) & Selling Exp.	₹2,50,000
Profit before Taxes	₹5,00,000
Less: Taxes	₹2,50,000
Net Profit	₹2,50,000

Balance Sheet

Particular	Rs, ₹	Particular	Rs, ₹
Equity share capital	20,00,000	Fixed Assets	54,00,000
10% Preference share capital	20,00,000	Stock	1,75,000
Reserves	10,00,000	Debtors	3,50,000
10% Debentures (Loan)	10,00,000	Bills receivable	50,000
Sundry creditors	1,00,000	Cash	2,25,000
Bank Overdraft Bills	1,50,000		
Accounts payable	45,000		
Outstanding expenses	5,000		
	63,00,000		63,00,000

Beside the details mentioned above, the opening stock was of ₹3,25,000, and debtors at the beginning was ₹3,00,000. Taking 360 days of the year, calculate the following ratios,

Net profit ratio (after tax).

Current ratio.

Debtors' turnover ratio.

4C) Green Energy Pvt. Ltd. is considering building a small solar power plant. The project requires an initial investment of \$1,200,000 and is expected to generate annual cash inflows for 6 years. The company uses a discount rate (MARR) of 8% per year. Because of uncertainty in solar energy production and market prices, management wants to perform a scenario analysis with three possible outcomes: (3)

Scenario	Probability	Annual Cash Inflows (\$)
Optimistic	30%	300,000
Most Likely	50%	250,000
Pessimistic	20%	180,000

Calculate the NPW of the project under each scenario. Determine the expected NPW of the project using the given probabilities under each scenario. Discuss the risk associated with the project and how

it could influence the investment decision.

8%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>
1	1.080	.9259	1.0000	1.0800	1.000	0.926	0
2	1.166	.8573	.4808	.5608	2.080	1.783	0.481
3	1.260	.7938	.3080	.3880	3.246	2.577	0.949
4	1.360	.7350	.2219	.3019	4.506	3.312	1.404
5	1.469	.6806	.1705	.2505	5.867	3.993	1.846
6	1.587	.6302	.1363	.2163	7.336	4.623	2.276
7	1.714	.5835	.1121	.1921	8.923	5.206	2.694
8	1.851	.5403	.0940	.1740	10.637	5.747	3.099
9	1.999	.5002	.0801	.1601	12.488	6.247	3.491
10	2.159	.4632	.0690	.1490	14.487	6.710	3.871

- 5A) Three types of drill bits can be used in a certain manufacturing operation. A bright high-speed steel (HSS) bit is the least expensive to buy, but it has a shorter life than either gold oxide or titanium nitride bits. The HSS bits will cost \$3,500 to buy and will last for 3 months under the conditions in which they will be used. The operating cost for these bits will be \$2,000 per month and salvage value is \$2,000. The gold oxide bits will cost \$6,500 to buy and will last for 6 months with an operating cost of \$1,500 per month and salvage value is \$4,000. (4)
The titanium nitride bits will cost \$7,000 to buy and will last 12 months with an operating cost of \$1,200 per month with salvage value of \$3,000.
At an interest rate of 15% per year, compounded monthly, which type of drill bit should be used based on Annual Worth analysis?

- 5B) A car purchased six years ago can be sold now for 3.5 lakhs, or can be kept for another five years. In that case, it will have an annual operation and maintenance cost of Rs. 50,000 for the first year of operation from now and increases every year by an amount of 20,000. It is expected to have a salvage value of Rs. 2.25 Lakhs, Rs. 2 Lakhs, Rs. 1.75 Lakhs, Rs. 1.5 Lakhs, Rs. 1.25 Lakhs for years seven to eleven respectively. Determine the economic service life of this old car if the interest rate is 12%.

12%

<i>n</i>	<i>F/P</i>	<i>P/F</i>	<i>A/F</i>	<i>A/P</i>	<i>F/A</i>	<i>P/A</i>	<i>A/G</i>	<i>P/G</i>
1	1.120	.8929	1.0000	1.1200	1.000	0.893	0	0
2	1.254	.7972	.4717	.5917	2.120	1.690	0.472	0.797
3	1.405	.7118	.2963	.4163	3.374	2.402	0.925	2.221
4	1.574	.6355	.2092	.3292	4.779	3.037	1.359	4.127
5	1.762	.5674	.1574	.2774	6.353	3.605	1.775	6.397
6	1.974	.5066	.1232	.2432	8.115	4.111	2.172	8.930
7	2.211	.4523	.0991	.2191	10.089	4.564	2.551	11.644
8	2.476	.4039	.0813	.2013	12.300	4.968	2.913	14.471
9	2.773	.3606	.0677	.1877	14.776	5.328	3.257	17.356
10	3.106	.3220	.0570	.1770	17.549	5.650	3.585	20.254

- 5C) Explain the difference between the Current Ratio and the Acid-Test Ratio, and discuss the significance of each. (2)

-----End-----

Event Name	Subject
058-END SEMSTER-ODD-2025	HUM_3021:ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

#	Course Outcome	Description	Marks
1	1	CO 1	8
2	2	CO 2	9

3	3	CO 3	3
4	4	CO 4	30
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	HUM_3021:ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	24(48%)
4	Analysis	7	26(52%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)